

NAME: _____

The Cookie Box Pricing Decision

a small-business pricing decision • Intro Finance

Amara owns a small bakery and wants to launch a line of gourmet cookie boxes. Her accountant is blunt: the new line only makes sense if it earns a **40% profit margin**, and it must carry \$1,200 a month of extra kitchen and lease costs on its own. Her business partner has already printed a price sheet using a rule of thumb. Before Amara commits, she wants the numbers checked.

Amara's numbers (everything you need)

- Cost to make one box: **\$6.00**. Fixed costs for the line: **\$1,200 per month**.
- Required profit margin: **40%**. Margin is profit as a share of the *price*.
- A campus market study estimates sales of about **400 boxes per month** at a price near \$10.
- The partner's price sheet: "cost plus a 40% markup, so $\$6.00 + 40\% = \mathbf{\$8.40}$ per box."

1. Check the Partner's Price

The partner insists a 40% markup gives a 40% margin. At \$8.40 per box, what margin does Amara actually earn? As a group, settle whether markup and margin are the same thing, and be ready to defend your answer. *Hint: Margin divides the profit by the price; markup divides it by the cost.*

2. Find the Right Price

What price gives a true 40% margin? Verify your price by computing its margin. *Hint: At the right price, the \$6.00 cost is exactly 60% of the price.*

3. Will the Line Survive?

For the partner's \$8.40 and for your Question 2 price: how many boxes per month does the line need just to cover the \$1,200 in fixed costs, and what is the monthly profit or loss at the expected 400 boxes? Which price makes the launch viable? *Hint: Each box contributes its profit toward the fixed costs; break-even is fixed costs divided by profit per box.*

As a group, write Amara a 2–3 sentence recommendation: launch or not, at what price, and what to tell the partner about the price sheet.

ANSWER KEY

1) Profit at \$8.40 is $8.40 - 6.00 = \$2.40$. $\text{Margin} = 2.40/8.40 = \boxed{28.6\%}$, well short of 40%. Markup and margin are different: markup measures against cost, margin against price, so a 40% markup always yields a smaller margin.

2) Require price $- 6.00 = 0.40 \times \text{price}$, so $6.00 = 0.60 \times \text{price}$ and price = $\boxed{\$10.00}$. Check: $(10.00 - 6.00)/10.00 = 40\%$.

3) Profit per box: \$2.40 at \$8.40; \$4.00 at \$10.00. Break-even: $1200/2.40 = \boxed{500 \text{ boxes}}$ at \$8.40 versus $1200/4.00 = \boxed{300 \text{ boxes}}$ at \$10.00. At the expected 400 boxes: \$8.40 loses $400(2.40) - 1200 = -\$240$ per month; \$10.00 earns $400(4.00) - 1200 = +\$400$ per month. Only the \$10.00 price is viable.

Verdict. Launch at \$10.00: expected demand (400) clears the 300-box break-even and nets about \$400 a month. At the partner's \$8.40 the line loses \$240 a month at the same demand, so the price sheet must be corrected before printing menus.